

Predicting PCL5 scores from tract metrics in the group with PTSD

Loading FD data...

FD → Total PCL

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.3757 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.981, p=0.579 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Total PCL

formula: PCL_SUM_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_SUM_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:46		
No. Iterations:	15		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0365	0.155	0.235	0.814	-0.268	0.341
UF_fd_mean_z	-0.1863	0.204	-0.914	0.361	-0.586	0.213
CB_fd_mean_z	0.0487	0.188	0.259	0.795	-0.319	0.416
ST_fd_mean_z	0.0571	0.197	0.290	0.772	-0.328	0.443
mri_age_z	-0.2378	0.191	-1.245	0.213	-0.612	0.136

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FD → Cluster Intrusion

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.3963 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.985, p=0.776 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Cluster Intrusion

formula: PCL_INTRU_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_INTRU_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:46		
No. Iterations:	15		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0113	0.148	0.077	0.939	-0.278	0.301
UF_fd_mean_z	-0.1200	0.194	-0.619	0.536	-0.500	0.260

CB_fd_mean_z	-0.1866	0.178	-1.046	0.296	-0.536	0.163
ST_fd_mean_z	0.1772	0.187	0.948	0.343	-0.189	0.544
mri_age_z	-0.2676	0.182	-1.474	0.140	-0.623	0.088

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Cluster Avoidance
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.9905 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.968, p=0.177 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Cluster Avoidance

formula: PCL_AVOID_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable: PCL_AVOID_z

No. Observations: 51

Model: RLM

Df Residuals: 46

Method: IRLS

Df Model: 4

Norm: HuberT

Scale Est.: mad

Cov Type: H1

Date: Mon, 21 Sep 2026

Time: 14:06:46

No. Iterations: 21

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0417	0.141	0.296	0.767	-0.234	0.318
UF_fd_mean_z	0.0407	0.185	0.220	0.826	-0.322	0.403
CB_fd_mean_z	0.1611	0.170	0.947	0.344	-0.172	0.495
ST_fd_mean_z	-0.1465	0.178	-0.821	0.411	-0.496	0.203
mri_age_z	-0.2373	0.173	-1.370	0.171	-0.577	0.102

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Cluster Cogn/Mood
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6409 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.959, p=0.074 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Cluster Cogn/Mood

formula: PCL_COMO_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results							
=====							
Dep. Variable:	PCL_COMO_z	No. Observations:	51				
Model:	RLM	Df Residuals:	46				
Method:	IRLS	Df Model:	4				
Norm:	HuberT						
Scale Est.:	mad						
Cov Type:	H1						
Date:	Mon, 21 Sep 2026						
Time:	14:06:46						
No. Iterations:	10						
=====							
	coef	std err	z	P> z	[0.025	0.975]	

Intercept	-0.0017	0.151	-0.011	0.991	-0.297	0.294	
UF_fd_mean_z	-0.2355	0.198	-1.190	0.234	-0.623	0.152	
CB_fd_mean_z	0.1126	0.182	0.618	0.536	-0.244	0.470	

```
ST_fd_mean_z 0.0521 0.191 0.273 0.785 -0.322 0.426
mri_age_z -0.0830 0.185 -0.448 0.654 -0.446 0.280
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If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Cluster Hyperarous
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4707 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.989, p=0.927 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Cluster Hyperarous

formula: PCL_HYPE_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

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=====
Dep. Variable:    PCL_HYPE_z  No. Observations:    51
Model:           RLM  Df Residuals:    46
Method:          IRLS  Df Model:        4
Norm:            HuberT
Scale Est.:      mad
Cov Type:        H1
Date:            Mon, 21 Sep 2026
Time:            14:06:47
No. Iterations:  13
=====
      coef  std err      z  P>|z|  [0.025  0.975]
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Intercept    0.0103    0.152    0.068  0.946  -0.288  0.308
UF_fd_mean_z -0.1400    0.200   -0.701  0.483  -0.532  0.252
CB_fd_mean_z  0.0650    0.184    0.353  0.724  -0.295  0.425
ST_fd_mean_z  0.0249    0.193    0.129  0.897  -0.353  0.403
mri_age_z    -0.2294    0.187   -1.226  0.220  -0.596  0.137
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```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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Loading FC data...
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FC → Total PCL
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```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1110 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.979, p=0.479 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Total PCL

formula: PCL_SUM_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

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=====
Dep. Variable:    PCL_SUM_z  No. Observations:    51
Model:           RLM  Df Residuals:    46
Method:          IRLS  Df Model:        4
Norm:            HuberT
Scale Est.:      mad
Cov Type:        H1
Date:            Mon, 21 Sep 2026
Time:            14:06:47
No. Iterations:  14
=====
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0401	0.156	0.257	0.797	-0.266	0.346
UF_fc_mean_res_z	0.0081	0.201	0.041	0.968	-0.385	0.402
CB_fc_mean_res_z	0.1020	0.216	0.473	0.636	-0.321	0.525
ST_fc_mean_res_z	0.1173	0.214	0.549	0.583	-0.302	0.536
mri_age_z	-0.2604	0.202	-1.287	0.198	-0.657	0.136

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Cluster Intrusion

Assumption checks

[1] Homoscedasticity
WARNING: Heteroscedasticity detected (Breusch-Pagan p=0.0447) — homoscedasticity not assumed. Not as critical for Huber regression

[2] Normality of residuals
Shapiro-Wilk W=0.980, p=0.526 — normality of residuals assumed.

[3] Multicollinearity (VIF)
No problematic multicollinearity detected.

Variable	VIF
UF_fc_mean_res_z	1.623417
CB_fc_mean_res_z	1.873909
ST_fc_mean_res_z	1.841847
mri_age_z	1.648634
Intercept	1.000000

[4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Cluster Intrusion

formula: PCL_INTRU_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_INTRU_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:47		
No. Iterations:	35		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0188	0.161	-0.116	0.907	-0.335	0.297
UF_fc_mean_res_z	0.1830	0.208	0.882	0.378	-0.224	0.590
CB_fc_mean_res_z	-0.1775	0.223	-0.796	0.426	-0.614	0.259
ST_fc_mean_res_z	0.0183	0.221	0.083	0.934	-0.415	0.452
mri_age_z	-0.2852	0.209	-1.364	0.173	-0.695	0.125

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Cluster Avoidance

Assumption checks

[1] Homoscedasticity
Breusch-Pagan p=0.2670 — homoscedasticity assumed.

[2] Normality of residuals
Shapiro-Wilk W=0.959, p=0.073 — normality of residuals assumed.

[3] Multicollinearity (VIF)
No problematic multicollinearity detected.

Variable	VIF
UF_fc_mean_res_z	1.623417
CB_fc_mean_res_z	1.873909
ST_fc_mean_res_z	1.841847
mri_age_z	1.648634
Intercept	1.000000

[4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Cluster Avoidance

formula: PCL_AVOID_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_AVOID_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:47		
No. Iterations:	17		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0589	0.142	0.415	0.678	-0.219	0.337
UF_fc_mean_res_z	0.1216	0.183	0.666	0.505	-0.236	0.479
CB_fc_mean_res_z	0.1067	0.196	0.544	0.586	-0.278	0.491
ST_fc_mean_res_z	0.3877	0.194	1.994	0.046	0.007	0.769
mri_age_z	-0.3927	0.184	-2.135	0.033	-0.753	-0.032

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Cluster Cogn/Mood

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2550 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.965, p=0.134 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Cluster Cogn/Mood

formula: PCL_COMO_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_COMO_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:48		
No. Iterations:	23		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0028	0.158	0.018	0.986	-0.307	0.313
UF_fc_mean_res_z	0.1179	0.204	0.579	0.562	-0.281	0.517
CB_fc_mean_res_z	0.0323	0.219	0.148	0.883	-0.396	0.461
ST_fc_mean_res_z	-0.0696	0.217	-0.321	0.748	-0.495	0.355
mri_age_z	0.0372	0.205	0.181	0.856	-0.365	0.439

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Cluster Hyperarous

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.5332 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.977, p=0.428 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Cluster Hyperarous

formula: PCL_HYPE_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	PCL_HYPE_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:06:48		
No. Iterations:	33		

	coef	std err	z	P> z	[0.025	0.975]
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Intercept      0.0100  0.147  0.068  0.946  -0.279  0.299
UF_fc_mean_res_z -0.1889  0.190 -0.995  0.320  -0.561  0.183
CB_fc_mean_res_z  0.1954  0.204  0.958  0.338  -0.204  0.595
ST_fc_mean_res_z  0.3048  0.202  1.508  0.132  -0.091  0.701
mri_age_z       -0.3973  0.191 -2.078  0.038  -0.772  -0.023
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```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

All models complete.

Predicting PID5 scores from tract metrics in the group with PTSD

Loading FD data...

FD → Negative Affectivity

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2166 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.990, p=0.940 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.618339
CB_fd_mean_z 1.400216
ST_fd_mean_z 1.480593
mri_age_z 1.442734
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Negative Affectivity

formula: Negative_Affectivity_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable: Negative_Affectivity_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:28
No. Iterations: 18

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0079	0.160	-0.049	0.961	-0.322	0.306
UF_fd_mean_z	-0.1161	0.206	-0.565	0.572	-0.519	0.287
CB_fd_mean_z	0.1487	0.191	0.777	0.437	-0.226	0.524
ST_fd_mean_z	0.0781	0.197	0.397	0.691	-0.308	0.464
mri_age_z	-0.2250	0.194	-1.158	0.247	-0.606	0.156

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FD → Detachment

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2564 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.970, p=0.241 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.618339
CB_fd_mean_z 1.400216
ST_fd_mean_z 1.480593
mri_age_z 1.442734
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Detachment

formula: Detachment_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable: Detachment_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:29
No. Iterations: 15

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0183	0.160	-0.114	0.909	-0.333	0.296
UF_fd_mean_z	-0.0276	0.206	-0.134	0.893	-0.432	0.376

```
CB_fd_mean_z 0.0666 0.192 0.347 0.728 -0.309 0.442
ST_fd_mean_z 0.0678 0.197 0.344 0.731 -0.319 0.454
mri_age_z 0.1239 0.195 0.636 0.525 -0.258 0.505
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```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Antagonism
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1891 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.907, p=0.001 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.618339
CB_fd_mean_z 1.400216
ST_fd_mean_z 1.480593
mri_age_z 1.442734
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Antagonism

formula: Antagonism_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

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Robust linear Model Regression Results
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Dep. Variable: Antagonism_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:29
No. Iterations: 17
=====
coef std err z P>|z| [0.025 0.975]
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Intercept -0.0958 0.103 -0.929 0.353 -0.298 0.106
UF_fd_mean_z -0.3341 0.133 -2.520 0.012 -0.594 -0.074
CB_fd_mean_z 0.3076 0.123 2.495 0.013 0.066 0.549
ST_fd_mean_z 0.2951 0.127 2.327 0.020 0.047 0.544
mri_age_z -0.0985 0.125 -0.787 0.431 -0.344 0.147
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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=====
FD → Disinhibition
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```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4004 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.968, p=0.198 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.618339
CB_fd_mean_z 1.400216
ST_fd_mean_z 1.480593
mri_age_z 1.442734
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Disinhibition

formula: Disinhibition_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable: Disinhibition_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:29
No. Iterations: 22
=====
coef std err z P>|z| [0.025 0.975]
-----
Intercept -0.0537 0.132 -0.407 0.684 -0.313 0.205
UF_fd_mean_z -0.2772 0.170 -1.632 0.103 -0.610 0.056
CB_fd_mean_z -0.0077 0.158 -0.049 0.961 -0.317 0.302
=====
```

```
ST_fd_mean_z 0.0332 0.162 0.205 0.838 -0.285 0.352
mri_age_z 0.1527 0.160 0.952 0.341 -0.162 0.467
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Psychoticism
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6979 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.976, p=0.385 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.618339
CB_fd_mean_z 1.400216
ST_fd_mean_z 1.480593
mri_age_z 1.442734
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Psychoticism

formula: Psychoticism_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable: Psychoticism_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:30
No. Iterations: 18
=====
coef std err z P>|z| [0.025 0.975]
-----
Intercept 0.0009 0.175 0.005 0.996 -0.343 0.345
UF_fd_mean_z -0.1532 0.225 -0.680 0.497 -0.595 0.288
CB_fd_mean_z -0.0602 0.210 -0.287 0.774 -0.471 0.351
ST_fd_mean_z 0.2351 0.216 1.091 0.275 -0.187 0.657
mri_age_z 0.1532 0.213 0.720 0.471 -0.264 0.570
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
=====
```

Loading FC data...

```
=====
FC → Negative Affectivity
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2568 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.989, p=0.914 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.663496
CB_fc_mean_res_z 1.873547
ST_fc_mean_res_z 1.700013
mri_age_z 1.533040
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Negative Affectivity

formula: Negative_Affectivity_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable: Negative_Affectivity_z No. Observations: 50
Model: RLM Df Residuals: 45
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:07:30
No. Iterations: 14
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0027	0.154	-0.018	0.986	-0.305	0.300
UF_fc_mean_res_z	-0.0810	0.201	-0.403	0.687	-0.475	0.313
CB_fc_mean_res_z	0.0236	0.213	0.111	0.912	-0.395	0.442
ST_fc_mean_res_z	-0.1562	0.203	-0.768	0.442	-0.555	0.242
mri_age_z	-0.1331	0.193	-0.689	0.491	-0.512	0.245

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Detachment

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.8727 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.976, p=0.405 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.663496
CB_fc_mean_res_z 1.873547
ST_fc_mean_res_z 1.700013
mri_age_z 1.533040
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Detachment

formula: Detachment_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Detachment_z	No. Observations:	50
Model:	RLM	Df Residuals:	45
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:07:30		
No. Iterations:	13		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0325	0.156	-0.208	0.835	-0.338	0.273
UF_fc_mean_res_z	-0.1610	0.203	-0.793	0.428	-0.559	0.237
CB_fc_mean_res_z	0.0279	0.215	0.130	0.897	-0.394	0.450
ST_fc_mean_res_z	-0.0235	0.205	-0.114	0.909	-0.426	0.379
mri_age_z	0.1121	0.195	0.576	0.565	-0.270	0.494

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Antagonism

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1718 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.925, p=0.004 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.663496
CB_fc_mean_res_z 1.873547
ST_fc_mean_res_z 1.700013
mri_age_z 1.533040
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Antagonism

formula: Antagonism_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Antagonism_z	No. Observations:	50
Model:	RLM	Df Residuals:	45
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:07:30		
No. Iterations:	17		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0956	0.108	-0.888	0.375	-0.307	0.116
UF_fc_mean_res_z	-0.2632	0.140	-1.875	0.061	-0.538	0.012
CB_fc_mean_res_z	0.3052	0.149	2.049	0.040	0.013	0.597
ST_fc_mean_res_z	-0.3545	0.142	-2.499	0.012	-0.633	-0.076
mri_age_z	0.0256	0.135	0.190	0.849	-0.238	0.290

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Disinhibition

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6291 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.973, p=0.298 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.663496
CB_fc_mean_res_z 1.873547
ST_fc_mean_res_z 1.700013
mri_age_z 1.533040
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Disinhibition

formula: Disinhibition_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Disinhibition_z	No. Observations:	50
Model:	RLM	Df Residuals:	45
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:07:31		
No. Iterations:	13		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0343	0.133	-0.259	0.796	-0.294	0.226
UF_fc_mean_res_z	-0.3872	0.173	-2.241	0.025	-0.726	-0.049
CB_fc_mean_res_z	0.2787	0.183	1.520	0.128	-0.081	0.638
ST_fc_mean_res_z	0.1238	0.175	0.709	0.478	-0.218	0.466
mri_age_z	0.1843	0.166	1.111	0.266	-0.141	0.509

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Psychoticism

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.3362 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.972, p=0.272 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.663496
CB_fc_mean_res_z 1.873547
ST_fc_mean_res_z 1.700013
mri_age_z 1.533040
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Psychoticism

formula: Psychoticism_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Psychoticism_z	No. Observations:	50
Model:	RLM	Df Residuals:	45
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:07:31		
No. Iterations:	19		

	coef	std err	z	P> z	[0.025	0.975]
--	------	---------	---	------	--------	--------

```
-----
Intercept      0.0093   0.170   0.055   0.956   -0.324   0.342
UF_fc_mean_res_z -0.0968  0.221  -0.438  0.662  -0.531  0.337
CB_fc_mean_res_z  0.0235  0.235   0.100  0.920  -0.437  0.484
ST_fc_mean_res_z -0.1977  0.224  -0.883  0.377  -0.636  0.241
mri_age_z       0.2258  0.212   1.063  0.288  -0.191  0.642
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

All models complete.

Predicting CTQ scores from tract metrics in the group with PTSD

Loading FD data...

FD → Total CTQ

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.7069 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.956, p=0.058 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Total CTQ

formula: CTQ_total_score_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQ_total_score_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:08:44		
No. Iterations:	20		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0626	0.149	0.420	0.674	-0.229	0.355
UF_fd_mean_z	0.0353	0.196	0.180	0.857	-0.348	0.419
CB_fd_mean_z	0.0535	0.180	0.297	0.766	-0.299	0.407
ST_fd_mean_z	-0.3659	0.189	-1.938	0.053	-0.736	0.004
mri_age_z	-0.0665	0.183	-0.363	0.717	-0.426	0.293

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FD → Emotional Abuse

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.7020 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.928, p=0.004 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Emotional Abuse

formula: CTQemotional_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQemotional_abuse_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:08:44		
No. Iterations:	20		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0697	0.128	0.544	0.586	-0.181	0.321
UF_fd_mean_z	0.0053	0.168	0.032	0.975	-0.324	0.335

```
CB_fd_mean_z 0.2316 0.155 1.497 0.135 -0.072 0.535
ST_fd_mean_z -0.2589 0.162 -1.596 0.111 -0.577 0.059
mri_age_z -0.0664 0.157 -0.421 0.673 -0.375 0.242
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Physical Abuse
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2705 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.951, p=0.036 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Physical Abuse

formula: CTQphysical_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable:  CTQphysical_abuse_z  No. Observations:      51
Model:          RLM  Df Residuals:          46
Method:         IRLS  Df Model:             4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:44
No. Iterations: 27
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0570   0.148  -0.384  0.701  -0.348   0.234
UF_fd_mean_z  0.0277   0.195   0.142  0.887  -0.354   0.410
CB_fd_mean_z  0.0540   0.179   0.301  0.763  -0.298   0.406
ST_fd_mean_z -0.3120   0.188  -1.660  0.097  -0.680   0.056
mri_age_z    0.0083   0.183   0.046  0.964  -0.349   0.366
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Sexual Abuse
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4506 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.984, p=0.723 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Sexual Abuse

formula: CTQsexual_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable:  CTQsexual_abuse_z  No. Observations:      51
Model:          RLM  Df Residuals:          46
Method:         IRLS  Df Model:             4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:45
No. Iterations: 22
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0022   0.146  -0.015  0.988  -0.288   0.283
UF_fd_mean_z  0.1150   0.191   0.601  0.548  -0.260   0.490
CB_fd_mean_z -0.1587   0.176  -0.901  0.368  -0.504   0.187
=====
```

```
ST_fd_mean_z -0.4168 0.185 -2.257 0.024 -0.779 -0.055
mri_age_z -0.3132 0.179 -1.747 0.081 -0.664 0.038
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Emotional Neglect
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.7906 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.929, p=0.004 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Emotional Neglect

formula: CTQemotional_neglect_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

```
=====
Dep. Variable:  CTQemotional_neglect_z  No. Observations:  51
Model:          RLM  Df Residuals:  46
Method:         IRLS  Df Model:  4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:45
No. Iterations:  22
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept  0.1130    0.137   0.824  0.410  -0.156  0.382
UF_fd_mean_z -0.1237   0.180  -0.687  0.492  -0.477  0.229
CB_fd_mean_z  0.2829   0.166   1.706  0.088  -0.042  0.608
ST_fd_mean_z -0.1754   0.174  -1.009  0.313  -0.516  0.165
mri_age_z   0.0139   0.169   0.082  0.934  -0.317  0.345
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Physical Neglect
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.9026 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.974, p=0.323 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 1.691171
CB_fd_mean_z 1.432794
ST_fd_mean_z 1.574300
mri_age_z 1.483711
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Physical Neglect

formula: CTQphys_neglect_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

```
=====
Dep. Variable:  CTQphys_neglect_z  No. Observations:  51
Model:          RLM  Df Residuals:  46
Method:         IRLS  Df Model:  4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:45
No. Iterations:  15
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0423   0.146  -0.290  0.772  -0.328  0.244
UF_fd_mean_z -0.0726   0.192  -0.378  0.705  -0.448  0.303
CB_fd_mean_z  0.0670   0.177   0.380  0.704  -0.279  0.413
ST_fd_mean_z -0.2179   0.185  -1.178  0.239  -0.581  0.145
=====
```

```
mri_age_z    -0.0254    0.180    -0.141    0.888    -0.377    0.327
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
=====
```

Loading FC data...

```
=====
FC -> Total CTQ
=====
```

Assumption checks

- ```

[1] Homoscedasticity
 Breusch-Pagan p=0.4804 — homoscedasticity assumed.

[2] Normality of residuals
 Shapiro-Wilk W=0.967, p=0.168 — normality of residuals assumed.

[3] Multicollinearity (VIF)
 No problematic multicollinearity detected.
 Variable VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000

[4] Linearity
 Inspect plot: residuals should form a random cloud around zero.
```

MLR results: FC -> Total CTQ

```

formula: CTQ_total_score_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z
```

#### Robust linear Model Regression Results

```
=====
Dep. Variable: CTQ_total_score_z No. Observations: 51
Model: RLM Df Residuals: 46
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 14:08:46
No. Iterations: 16
=====
 coef std err z P>|z| [0.025 0.975]

Intercept 0.0550 0.132 0.416 0.678 -0.204 0.314
UF_fc_mean_res_z 0.3348 0.170 1.967 0.049 0.001 0.668
CB_fc_mean_res_z -0.2267 0.183 -1.240 0.215 -0.585 0.132
ST_fc_mean_res_z 0.1517 0.181 0.837 0.403 -0.204 0.507
mri_age_z 0.0041 0.171 0.024 0.981 -0.332 0.340
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FC -> Emotional Abuse
=====
```

Assumption checks

- ```
-----
[1] Homoscedasticity
    Breusch-Pagan p=0.8038 — homoscedasticity assumed.

[2] Normality of residuals
    WARNING: Shapiro-Wilk W=0.944, p=0.017 — residuals not normally distributed! (Not a formal requirement for Huber regression)

[3] Multicollinearity (VIF)
    No problematic multicollinearity detected.
    Variable    VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000

[4] Linearity
    Inspect plot: residuals should form a random cloud around zero.
```

MLR results: FC -> Emotional Abuse

```
-----
formula: CTQemotional_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z
```

Robust linear Model Regression Results

```
=====
Dep. Variable:  CTQemotional_abuse_z  No. Observations:      51
Model:          RLM  Df Residuals:      46
Method:         IRLS  Df Model:         4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:46
No. Iterations: 30
=====
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.0778	0.141	0.551	0.582	-0.199	0.355
UF_fc_mean_res_z	0.2132	0.182	1.173	0.241	-0.143	0.569
CB_fc_mean_res_z	-0.1029	0.195	-0.527	0.598	-0.486	0.280
ST_fc_mean_res_z	-0.1219	0.194	-0.629	0.529	-0.501	0.258
mri_age_z	0.1156	0.183	0.631	0.528	-0.243	0.475

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Physical Abuse

Assumption checks

[1] Homoscedasticity

Breusch-Pagan p=0.5876 — homoscedasticity assumed.

[2] Normality of residuals

WARNING: Shapiro-Wilk W=0.921, p=0.002 — residuals not normally distributed! (Not a formal requirement for Huber regression)

[3] Multicollinearity (VIF)

No problematic multicollinearity detected.

Variable VIF

UF_fc_mean_res_z 1.623417

CB_fc_mean_res_z 1.873909

ST_fc_mean_res_z 1.841847

mri_age_z 1.648634

Intercept 1.000000

[4] Linearity

Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Physical Abuse

formula: CTQphysical_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQphysical_abuse_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:08:46		
No. Iterations:	13		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0377	0.138	-0.273	0.785	-0.308	0.233
UF_fc_mean_res_z	0.0961	0.178	0.541	0.589	-0.252	0.444
CB_fc_mean_res_z	0.1497	0.191	0.784	0.433	-0.224	0.524
ST_fc_mean_res_z	0.2942	0.189	1.554	0.120	-0.077	0.665
mri_age_z	-0.0353	0.179	-0.197	0.844	-0.386	0.316

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Sexual Abuse

Assumption checks

[1] Homoscedasticity

Breusch-Pagan p=0.7120 — homoscedasticity assumed.

[2] Normality of residuals

Shapiro-Wilk W=0.978, p=0.459 — normality of residuals assumed.

[3] Multicollinearity (VIF)

No problematic multicollinearity detected.

Variable VIF

UF_fc_mean_res_z 1.623417

CB_fc_mean_res_z 1.873909

ST_fc_mean_res_z 1.841847

mri_age_z 1.648634

Intercept 1.000000

[4] Linearity

Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Sexual Abuse

formula: CTQsexual_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQsexual_abuse_z	No. Observations:	51
Model:	RLM	Df Residuals:	46
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:08:46		
No. Iterations:	19		

	coef	std err	z	P> z	[0.025	0.975]
--	------	---------	---	------	--------	--------

```

=====
Intercept      0.0021    0.143    0.014    0.989   -0.278    0.282
UF_fc_mean_res_z  0.3147    0.184    1.713    0.087   -0.045    0.675
CB_fc_mean_res_z -0.4051    0.197   -2.052    0.040   -0.792   -0.018
ST_fc_mean_res_z  0.2488    0.196    1.272    0.203   -0.135    0.632
mri_age_z       -0.2282    0.185   -1.233    0.218   -0.591    0.135
=====

```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```

=====
FC → Emotional Neglect
=====

```

Assumption checks

```

=====
[1] Homoscedasticity
    Breusch-Pagan p=0.7724 — homoscedasticity assumed.

[2] Normality of residuals
    WARNING: Shapiro-Wilk W=0.919, p=0.002 — residuals not normally distributed! (Not a formal requirement for Huber regression)
/home/amygdala/miniconda3/envs/project/lib/python3.11/site-packages/statsmodels/graphics/utls.py:55: RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot
interface ('matplotlib.pyplot.figure') are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam 'figure.max_open_warning'). Consider using
'matplotlib.pyplot.close()'.
  fig = plt.figure()

[3] Multicollinearity (VIF)
    No problematic multicollinearity detected.
    Variable    VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000

[4] Linearity
    Inspect plot: residuals should form a random cloud around zero.
=====

```

MLR results: FC → Emotional Neglect

formula: CTQemotional_neglect_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

```

=====
Dep. Variable:  CTQemotional_neglect_z  No. Observations:      51
Model:          RLM  Df Residuals:        46
Method:         IRLS  Df Model:           4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:47
No. Iterations: 20
=====

```

```

=====
      coef    std err          z      P>|z|    [0.025    0.975]
-----
Intercept    0.0945    0.128     0.738    0.461   -0.157    0.346
UF_fc_mean_res_z  0.2605    0.165     1.580    0.114   -0.063    0.583
CB_fc_mean_res_z -0.2081    0.177    -1.175    0.240   -0.555    0.139
ST_fc_mean_res_z -0.1153    0.176    -0.657    0.511   -0.459    0.229
mri_age_z      0.2247    0.166     1.353    0.176   -0.101    0.550
=====

```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```

=====
FC → Physical Neglect
=====

```

Assumption checks

```

=====
[1] Homoscedasticity
    Breusch-Pagan p=0.8315 — homoscedasticity assumed.

[2] Normality of residuals
    Shapiro-Wilk W=0.983, p=0.661 — normality of residuals assumed.

[3] Multicollinearity (VIF)
    No problematic multicollinearity detected.
    Variable    VIF
UF_fc_mean_res_z 1.623417
CB_fc_mean_res_z 1.873909
ST_fc_mean_res_z 1.841847
mri_age_z 1.648634
Intercept 1.000000

[4] Linearity
    Inspect plot: residuals should form a random cloud around zero.
=====

```

MLR results: FC → Physical Neglect

formula: CTQphys_neglect_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

```

=====
Dep. Variable:  CTQphys_neglect_z  No. Observations:      51
Model:          RLM  Df Residuals:        46
Method:         IRLS  Df Model:           4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:08:47
=====

```

No. Iterations: 15

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0171	0.152	-0.112	0.910	-0.314	0.280
UF_fc_mean_res_z	0.2863	0.195	1.467	0.142	-0.096	0.669
CB_fc_mean_res_z	-0.0858	0.210	-0.409	0.682	-0.497	0.325
ST_fc_mean_res_z	0.0747	0.208	0.360	0.719	-0.333	0.482
mri_age_z	0.1133	0.197	0.576	0.564	-0.272	0.499

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

All models complete.